

When the vessel breaks

Pathophysiology of non-traumatic intracerebral haemorrhage. Chapter 5, Oxford Textbook of Stroke and Cerebrovascular Disease (ed. Norrving, OUP 2014), by Constanza Rossi and Charlotte Cordonnier. Figures come from the primary studies listed overleaf.

1 The vessel decides where it bleeds.

2 The first hours decide how big it gets.

3 The blood keeps injuring the brain for days.

01 The vessel

About 80–85% of ICH is **primary**: a small vessel weakened by chronic disease ruptures. About 15–20% is **secondary** to an identifiable lesion or condition.

The two small-vessel diseases

	HYPERTENSIVE	AMYLOID (CAA)
Vessels	Deep perforators	Cortical, leptomeningeal
Wall	Lipohyalinosis, fibrinoid necrosis; microaneurysms (debated)	β-amyloid replaces smooth muscle
Bleed site	Putamen, thalamus, pons, cerebellum	Lobar
Microbleeds	Deep	Strictly lobar
Patient	Long-standing hypertension	Age 55+; APOE ε2/ε4

Causes in one cohort (SMASH-U, n = 1013)

Hypertension	35%
Undetermined	21%
Amyloid angiopathy	20%
Anticoagulants (largest bleeds; 54% dead by 3 months)	14%
Structural lesion (AVM, cavernoma)	5%
Systemic disease (cirrhosis, low platelets)	5%

Microbleeds (tiny old leaks, black dots on blood-sensitive MRI) appear in 5% of healthy adults, 34% after ischaemic stroke and 60% of people with ICH.

The patient in the talk. Jill Bolte Taylor was 37 and normotensive when an AVM in her left hemisphere ruptured on 10 December 1996. It was a textbook secondary ICH.

Secondary causes: always ask

-  **Vascular malformations:** AVM, cavernoma, dural fistula, aneurysm. Commonest in younger patients.
-  **Antithrombotic drugs:** make bleeds larger and keep them growing for longer.
-  **Tumours:** primary or metastatic; fragile new vessels.
-  **Cerebral venous thrombosis:** blocked drainage raises venous pressure.
-  **Haemorrhagic transformation** of an infarct.
-  **Others:** cocaine, amphetamines, vasculitis, RCVS, moyamoya, coagulopathy.

Red flags for a hidden lesion

- ✓ Aged 45 or under, or no history of hypertension
- ✓ Lobar bleed, or blood only in the ventricles
- ✓ CT clues: subarachnoid blood, calcification, abnormal vessels
- ✓ A bleed next to a venous sinus, or a dense sinus
- ✓ On anticoagulants: check clotting at once

65%

had a lesion on angiography: aged 45 or under, normotensive, lobar bleed

0%

had one: over 45, hypertensive, deep bleed

Modified Boston criteria for CAA

Definite	Full post-mortem shows severe CAA with a lobar bleed.
Probable	Age 55+; multiple lobar or cortical bleeds (or one plus cortical superficial siderosis); no other cause.
Possible	Age 55+; a single lobar bleed or siderosis; no other cause.

02 The first hours

01 Rupture. A diseased small artery gives way under arterial pressure.

03 Avalanche. The growing clot tears vessels at its edge (Fisher, 1971).

02 Haematoma. Blood splits the tissue planes.

04 Tamponade. Tissue pressure and clotting stop the bleed.

38 in 100

grew by over a third within 20 h (26 within the first hour)

+5%

hazard of death for every 10% of growth

⌚ Higher risk of growth

Early presentation, a large baseline bleed, anticoagulant use and a **spot sign** (contrast leaking into the clot on CT angiography).

○ Size in context

A golf ball holds about 40 mL. Volume of 60 mL or more with a GCS of 8 or less predicted 91% mortality at 30 days. A clot that grows by a third is only about 10% wider on a scan, so growth is easy to miss by eye.

≈45%

extend into the ventricles: hydrocephalus risk and a worse outcome

40%

median case fatality at 1 month, unchanged 1980–2008

03 The days after

WHEN	WHAT DRIVES THE OEDEMA
Hours	Clot retraction squeezes serum into the tissue
Days 1–2	The coagulation cascade makes thrombin, which opens the blood–brain barrier
Day 3+	Red cells lyse; haemoglobin, haem and iron drive oxidative injury
Weeks	Macrophages clear the clot; a cavity and haemosiderin remain

Oedema grew by about 75% in the first 24 hours in one study, and it can keep building into the second week.

🔥 Three toxins, one result

⚗️ **Thrombin:** needed to stop the bleed, toxic at high levels.

⊗ **Iron:** released from lysed red cells; fuels free radicals.

🔥 **Inflammation:** microglia within hours, then neutrophils, macrophages, cytokines and MMP-9.

All three lead to a leaky barrier, more oedema and neuronal death. Most of this evidence comes from animal models.

Myth: the rim around the clot is starving tissue. PET at 5–22 h showed blood flow at 56% of the other side, but oxygen use at 48%. The tissue is resting, not starving. In ICH ADAPT, lowering systolic pressure below 150 mmHg did not reduce flow around the clot.

04 At the bedside: every stage is a window

BEFORE AND AFTER

The vessel

Control blood pressure; it is the strongest tool against a first or second bleed. Review antithrombotics. Older adult with a lobar bleed: think CAA.

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FIRST 24 HOURS

The first hours

Image fast, lower blood pressure early and reverse anticoagulants urgently. Keep neuro observations close; a falling GCS needs action now.

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DAYS TO WEEKS

The days after

Expect fluctuation as oedema peaks. New drowsiness, headache, vomiting or a new deficit: stop and escalate. Reassess before every session and pace therapy.